

Solutions to JEE Main - 10 | JEE-2022

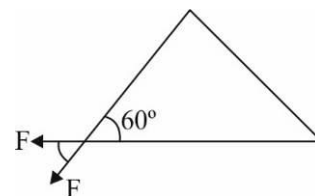
PHYSICS

SECTION-1

1.(B) Let $F = \frac{Q^2}{4\pi\epsilon_0 a^2}$

Then, the net electrostatic force acting on any one of the particles is

$$F_{net} = \sqrt{F^2 + F^2 + 2F^2 \cos 60^\circ} = \sqrt{3}F = \frac{\sqrt{3}Q^2}{4\pi\epsilon_0 a^2}$$



2.(B) In the balanced condition, i.e. when the current through the central branch is zero, $\frac{R_1}{R_2} = \frac{R_{AJ}}{R_{JB}}$

3.(C) Current in the circuit, $i = \frac{12+12}{10+1+1} = 2 \text{ A}$ P.D. across $E_1 = 12 - i(1) = 10 \text{ V}$

4.(C) Velocity electrons is much greater than the drift velocity. The electrons move in a random manner at very high velocities and the drift velocity is just the average rate at which the electrons drift through the conductor.

The number of electrons passing through a cross-section of the conductor in a given time interval is nAv_D

Here, n is the number of electrons per unit volume, A is the cross-section area and v_D is the drift velocity

5.(B) Taking a Gaussian surface of radius x (such that $x > R$), $E(4\pi x^2) = \frac{Q_{encl}}{\epsilon_0}$

Now, $Q_{encl} = \left(\frac{4\pi}{3} \left(\frac{R}{2} \right)^3 \right) (2\rho) + \left(\frac{4\pi}{3} \left(R^3 - \left(\frac{R}{2} \right)^3 \right) \right) (\rho) = \frac{3\pi\rho R^3}{2}$

Therefore, $E = \frac{3\rho R^3}{8\epsilon_0 x^2}$

Now, potential at the surface of the sphere is $V = -\int_{\infty}^R E dx = \frac{3\rho R^2}{8\epsilon_0}$

6.(D) We know that the electric field at a point P due to an infinitely long charged wire is $E = \frac{\lambda}{2\pi\epsilon_0 r}$

Here, r is the perpendicular distance of the point P from the wire

The field is directed perpendicular to the wire, away from the wire if the charge is positive and towards the wire if the charge is negative

Hence, the net field at the point $(2a, 0)$ is: $E = \frac{\lambda}{2\pi\epsilon_0 (3a)} \hat{i} + \frac{\lambda}{2\pi\epsilon_0 a} (-\hat{i}) = \frac{\lambda}{3\pi\epsilon_0 a} (-\hat{i})$

7.(A) Current through the main circuit, $i = \frac{24}{10+2} = 2 \text{ A}$

Then, emf of cell E_2 is equal to the potential difference across PJ, i.e.

$$E_2 = V_{PJ} = \left(\frac{70}{120} \right) R_{PQ} i = \left(\frac{70}{120} \right) (10)(2) = 11.7 \text{ V}$$

- 8.(A)** Due to cylindrical symmetry, the electric field at all points will be radially outward.
For $r < R$, let us define a Gaussian surface which is a closed cylinder of length L and radius r
Then, applying Gauss law,

$$[E(r)](2\pi rL) = \frac{\rho(\pi r^2 L)}{\epsilon_0} \Rightarrow E(r) = \frac{\rho r}{2\epsilon_0} \quad \text{if } r < R$$

Similarly, for $r > R$, defining a Gaussian surface which is a closed cylinder of length L and radius r and applying Gauss law,

$$[E(r)](2\pi rL) = \frac{\rho(\pi R^2 L)}{\epsilon_0} \Rightarrow E(r) = \frac{\rho R^2}{2\epsilon_0 r} \quad \text{if } r > R$$

$$\text{Now, for some } a < R, \text{ and some } b > R, E(a) = E(b) \Rightarrow \frac{\rho a}{2\epsilon_0} = \frac{\rho R^2}{2\epsilon_0 b} \Rightarrow ab = R^2$$

- 9.(A)** Let $\vec{E} = E_x \hat{i} + E_y \hat{j}$

Work done by electric field in journey $O \rightarrow A$,

$$W_1 = q\vec{E} \cdot (\hat{i}) = qE_x \Rightarrow E_x = \frac{W_1}{q} = 2000 \text{ V/m}$$

Work done by electric field in journey $A \rightarrow B$,

$$W_2 = q\vec{E} \cdot (-\hat{j}) = -qE_y \Rightarrow E_y = -\frac{W_2}{q} = -1500 \text{ V/m}$$

$$\text{So, } \vec{E} = 2000\hat{i} + 1500(-\hat{j}) = 2500\left(\frac{4\hat{i} - 3\hat{j}}{5}\right) \text{ V/m}$$

- 10.(D)** Let the emf of the battery be V

$$\text{We know that } H_1 = \frac{V^2}{R_1 + R_2} \quad \text{and} \quad H_2 = \frac{V^2}{\left(\frac{R_1 R_2}{R_1 + R_2}\right)} = \frac{(R_1 + R_2)V^2}{R_1 R_2}$$

$$\text{So, } \frac{H_1}{H_2} = \frac{R_1 R_2}{(R_1 + R_2)^2} = \frac{4}{25} \Rightarrow \frac{\left(\frac{R_1}{R_2}\right)}{\left(\frac{R_1}{R_2} + 1\right)^2} = \frac{4}{25}$$

$$\text{Let } \frac{R_1}{R_2} = x \quad \text{Then, } 4(1+x)^2 = 25x \quad \text{Solving, we get } x = \frac{1}{4}$$

- 11.(B)** Resolving the dipole into two parts: $p\hat{i} = \left(\frac{p}{\sqrt{2}}\right)\left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right) + \left(\frac{p}{\sqrt{2}}\right)\left(\frac{\hat{i} - \hat{j}}{\sqrt{2}}\right)$

Now, the point (a, a) lies on the axis of the first dipole and the equator of the second dipole

Therefore, the electric field at the point (a, a) is

$$\vec{E} = \left(\frac{2}{4\pi\epsilon_0(\sqrt{2}a)^3}\right)\left(\frac{p}{\sqrt{2}}\right)\left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right) + \left(\frac{-1}{4\pi\epsilon_0(\sqrt{2}a)^3}\right)\left(\frac{p}{\sqrt{2}}\right)\left(\frac{\hat{i} - \hat{j}}{\sqrt{2}}\right); \quad \vec{E} = \left(\frac{\sqrt{5}}{16}\right)\left(\frac{p}{\pi\epsilon_0 a^3}\right)\left(\frac{\hat{i} + 3\hat{j}}{\sqrt{10}}\right)$$

- 12.(C)** Let the value of the shunt resistance be S

$$\text{Then, } i_G G = (i - i_G)S \Rightarrow S = \frac{i_G G}{i - i_G} = \frac{(0.1)(10)}{10 - 0.1} = \frac{10}{99} \Omega$$

$$\text{Net resistance of the galvanometer, } R_{\text{net}} = \frac{GS}{G + S} = \frac{(10)\left(\frac{10}{99}\right)}{10 + \frac{10}{99}} = 0.1 \Omega$$

- 13.(A)** A dipole is composed of a positive and a negative charge, and the dipole moment vector points from the negative charge towards the positive charge

If the ring is positively charged, then it is simple to see that the electrostatic potential energy of the system increases by rotation of the dipole. This is because initially, the negative charge of the dipole was as close as possible to the ring and the positive charge of the dipole was as far away from the ring. This is a stable equilibrium, since out of all possible rotational orientations of the dipole, this position has the least potential energy.

Since the work done is equal to the change in potential energy, work done is positive if the ring is positively charged.

By a similar argument, we can see that if the ring is negatively charged, the work done is negative

- 14.(A)** Dipole moment of the dipole initially, $\vec{p}_i = QL(-\hat{i})$

And after counter-clockwise rotation by 60° , the dipole moment becomes

$$\vec{p}_f = QL\left(-\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$$

We know that

Work done in rotation = Change in potential energy of dipole

$$\text{Therefore, the work done, } W = \left(-\vec{p}_f \cdot (E(-\hat{i}))\right) - \left(-\vec{p}_i \cdot (E(-\hat{i}))\right) = \left(-\frac{QEL}{2}\right) - (-QEL) = \frac{QEL}{2}$$

Alternative Solution

Instead of looking at the motion as a rotation, we can just see it as the displacement of these charges from their initial position to their final position. We need to be concerned with only their displacement as the path taken does not affect the work done.

If a charged particle is moved slowly in a uniform electric field, such that its displacement is $\vec{\Delta r}$, the work done by external force is $-(q\vec{E}) \cdot \vec{\Delta r}$ (while the work done by the electrostatic force is $(q\vec{E}) \cdot \vec{\Delta r}$)

Since here the electric field is along the X direction, only the X-component of displacement is needed (because of the dot product)

The X-displacements of the charges $+Q$ and $-Q$ due to counter-clockwise rotation by 60° are respectively

$$\frac{L}{2}(1 - \cos 60^\circ) = \frac{L}{4} \quad \text{and} \quad -\frac{L}{2}(1 - \cos 60^\circ) = -\frac{L}{4}$$

$$\text{Therefore, work done on the charge } +Q, \quad W_+ = -(Q)(-E)\left(\frac{L}{4}\right) = \frac{QEL}{4}$$

$$\text{And, work done on the charge } -Q, \quad W_- = -(-Q)(-E)\left(-\frac{L}{4}\right) = \frac{QEL}{4}$$

$$\text{So, total work done, } W = W_+ + W_- = \frac{QEL}{2}$$

- 15.(B)** The magnitude of the change in velocity is $\Delta v = \sqrt{v^2 + (\sqrt{3}v)^2} = 2v$

This magnitude of change in velocity must be equal to the acceleration multiplied by the time (this works because the acceleration is constant)

$$\text{So, } 2v = \left(\frac{qE}{m}\right)t_0 \Rightarrow E = \frac{2mv}{qt_0}$$

- 16.(B)** An ideal voltmeter has an infinite resistance, and hence no current will flow through it. So, a clockwise current will flow through the outer loop. This current,

$$I = \frac{20-12}{10} = 0.8 \text{ A}$$

Now, $V_p - V_Q = (0.8)(2) + 12 + (0.8)(4) = 16.8 \text{ V}$

Therefore, the voltmeter reads 16.8 V

- 17.(C)** Let the emf of the battery be E

Then, $I_0 = \frac{E}{20+1} = \frac{E}{21}$

Now, after the cutting, the resistance of each piece of the wire is 5Ω

Therefore, the net resistance of the parallel combination of all 4 of them is $\frac{5}{4} \Omega$

So, the current through the battery is now $I = \frac{E}{\frac{5}{4}+1} = \frac{4E}{9} = \frac{4}{9}(21I_0) = \frac{28}{3}I_0$

- 18.(A)** The potential difference between the shells depends on the electric field in the space between them, i.e. if this electric field is $\vec{E}$, then the potential difference between the shells is

$$V_{2R} - V_R = -\int_R^{2R} \vec{E} \cdot d\vec{r}$$

But this electric field $\vec{E}$ depends only on Q_1 , and not on Q_2 (we can see this simply by applying Gauss law on a spherical surface of radius r , such that $R < r < 2R$), and hence the potential difference depends only on Q_1 , and not on Q_2

The electric field in the region $r > 2R$ is proportional to $Q_1 + Q_2$. Again, this can be seen by applying Gauss law on a spherical surface of radius $r > 2R$

- 19.(B)** Since the field is uniform, the net flux through the hemispherical surface (say Φ_{HS}) and the flux through the circular base (say Φ_{CB}) must be related as:

$$\Phi_{HS} + \Phi_{CB} = 0$$

(with the convention that flux going out from the interior of the hemisphere is taken positive and flux entering the hemisphere from the exterior is taken negative)

So, Φ_{HS} and Φ_{CB} have the same magnitude

But, we know that the flux through the circular base, i.e. Φ_{CB} , depends only on the component of electric field perpendicular to the base, i.e. the Z component. Hence, Φ_{HS} will also depend only on the Z component of the field

- 20.(D)** We can find the electric field at any point by superposition, i.e. by finding the field at the point due to each plate separately and then performing a vector addition

Therefore, taking rightward as positive,

$$E_A = -\frac{\sigma}{2\epsilon_0} + \left(-\frac{2\sigma}{2\epsilon_0}\right) = -\frac{3\sigma}{2\epsilon_0}$$

$$E_B = \frac{\sigma}{2\epsilon_0} + \left(-\frac{2\sigma}{2\epsilon_0}\right) = -\frac{\sigma}{2\epsilon_0}$$

So, $\vec{E}_A = 3\vec{E}_B$

SECTION-2

- 21.(7) If the current through the ammeter branch is zero, then the potential difference across this branch must simply be equal to the emf of the battery in this branch, i.e. 12 V

Therefore, the p.d. across the $6\ \Omega$ resistance is also 12 V, and hence the current through it is

$$\frac{12}{6} = 2\text{ A}$$

Now, since the current through the ammeter branch is zero, we know by using KCL at any of the two junctions that the 2 A current flow only in the top-right loop

$$\text{So, } \frac{30}{R+2+6} = 2 \Rightarrow R = 7\ \Omega$$

- 22.(9) The electric field in the wire, $E = \frac{V}{L}$

$$\text{And, the current density in the wire, } j = \frac{E}{\rho} = \frac{V}{\rho L} = \frac{100}{(10^{-7})(1)} = 10^9\text{ A/m}^2$$

- 23.(2) The total energy (kinetic energy + electrostatic potential energy) of the moving charge is conserved. So, its kinetic energy is maximum when its potential energy is minimum, and this happens when it is at the origin

$$\text{Conserving energy, } 2 \left(\frac{(Q)(-Q)}{4\pi\epsilon_0 \sqrt{a^2 + (\sqrt{3}a)^2}} \right) = 2 \left(-\frac{(Q)(-Q)}{4\pi\epsilon_0 a} \right) + \frac{1}{2}mv^2 \Rightarrow v = \left(\frac{Q^2}{2\pi\epsilon_0 a m} \right)^{1/2}$$

- 24.(5) Since the electrostatic force acting on the bob is constant in magnitude and direction, and therefore is just like gravity, the switching on of the electric field results in changing the effective value of gravity

Usually, the value of acceleration due to gravity is $g = \frac{\text{Net force due to gravity}}{\text{mass}}$

So, we can define the effective value of gravity as

$$g_{\text{EFF}} = \frac{\text{Net force due to gravity and electric field}}{\text{mass}}$$

$$\text{Therefore, } g_{\text{EFF}} = \frac{mg - qE}{m} = g - \frac{qE}{m}$$

$$g_{\text{EFF}} = g - \frac{qE}{m} = 10 - \frac{(10^{-3})(360)}{0.1} = 6.4\text{ m/s}^2$$

If the time period was initially $T_0 = 2\pi\sqrt{\frac{L}{g}}$, it now becomes $T = 2\pi\sqrt{\frac{L}{g_{\text{EFF}}}}$

$$\text{Since } \frac{g_{\text{EFF}}}{g} = \frac{64}{100}, \quad \frac{T}{T_0} = \sqrt{\frac{g}{g_{\text{EFF}}}} = \sqrt{\frac{100}{64}} = \frac{5}{4}$$

- 25.(4) (a) $4\ \Omega$ and $6\ \Omega$ are in series $\Rightarrow 10\ \Omega$
 (b) Now, $10\ \Omega$ and $10\ \Omega$ are in parallel $\Rightarrow 5\ \Omega$
 (c) Now, $5\ \Omega$ and $3\ \Omega$ are in series $\Rightarrow 8\ \Omega$
 (d) Finally, $8\ \Omega$ and $2\ \Omega$ are in parallel $\Rightarrow 1.6\ \Omega$

$$\text{So, current through the battery, } I = \frac{15}{1.6+1.4} = 5\text{ A}$$

$$\text{Therefore, potential difference between B and C, } V_{BC} = (5)(1.6) = 8\text{ V}$$

$$\text{So, current through } 2\ \Omega, \quad I_{2\Omega} = \frac{8}{2} = 4\text{ A}$$

Chemistry

SECTION-1

- 1.(D) $\Delta T_b = iK_b m = 1 \times K_b \times \frac{1000}{(1200-180)} = 0.98 K_b$
- 2.(B) Since, CaCl_2 dissociates in the aqueous solution, hence the molecular weight of CaCl_2 is less than 111.
- 3.(D) Theory based
- 4.(C) $Y_A = \frac{P_A}{P} \Rightarrow \frac{P_A^0 X_A}{P}$; $Y_B = \frac{P_B^0 X_B}{P}$; $\frac{Y_A}{Y_B} = \frac{P_A^0}{P_B^0} \times \frac{X_A}{X_B}$ $\therefore \frac{Y_A}{Y_B} > \frac{X_A}{X_B}$
- 5.(C) $\frac{P_A^0 - P}{P} = \frac{i n_B}{n_A}$
 $\frac{20.0108 - 20}{20} = \frac{3 \times n_B}{n_A}$; $\frac{0.0108}{60} = \frac{n_B}{n_A}$
 Let no. of moles of solvent = 1
 $\therefore$ Moles of solute = $0.0108/60$
 So molality = $\frac{0.0108/60}{18 \times 10^{-3}} = 10^{-2}$
 For very dilute aqueous solution,
 Molarity = Molality $\therefore K_{sp} = 4s^3 = 4 \times (10^{-2})^3 = 4 \times 10^{-6}$
- 6.(B) $\Delta T_f = K_f \cdot m \cdot i$
 $0.2046 = 1.86 \times 0.1(1 + \alpha)$
 $\alpha = 0.1$
 $\text{HCOOH(aq)} \rightleftharpoons \text{H}^+(\text{aq}) + \text{OH}^-(\text{aq})$
 $K_a = \frac{C\alpha^2}{(1-\alpha)} = \frac{0.1 \times (0.1)^2}{0.9} = \frac{1}{9} \times 10^{-2}$
 $K_h = \frac{K_w}{K_a} = \frac{10^{-14}}{\frac{1}{9} \times 10^{-2}} \Rightarrow 9 \times 10^{-12}$
- 7.(D) Maximum solubility in 50 gm = 18 gm
 Extra solute will not affect vapour pressure
- 8.(A) Benzene more volatile than toluene
 Hence from figure
 Statement (3) is correct $X_{\text{toluene}} = 0.7$ and $Y_{\text{toluene}} = 0.4$
 $\Rightarrow X_{\text{benzene}} = 0.3$ and $Y_{\text{toluene}} = 0.4$
 Statement (4) is correct because $X_{\text{benzene}} = 0.7$ { $Y_{\text{benzene}} > 0.7$ }
 $\Rightarrow Y_{\text{toluene}}$ necessarily < 0.3
 Statement (1) incorrect: because 4 option is correct
 Statement (2) is correct: $X_{\text{toluene}} = 0.3 \Rightarrow X_{\text{benzene}} = 0.7$
 $\Rightarrow Y_{\text{benzene}}$ must be > 0.7

9.(C) $T_b - T_f = 105$

$$(100 + \Delta T_b) - (-\Delta T_f) = 105$$

$$\Delta T_b + \Delta T_f = 5$$

$$(k_b + k_f)m = 5$$

$$2.37m = 5 ; m = 2.1$$

$$2.1 = \frac{\left(\frac{w}{342}\right)}{0.1}; w = 71.82 \text{ g}$$

$$w \approx 72$$

10.(C) $\ln \frac{P_2}{P_1} = \frac{\Delta H}{R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right)$

$$\ln \frac{2}{1} = \frac{200}{2} \left[\frac{1}{100} - \frac{1}{T_2} \right]$$

$$T_2 = \frac{100}{0.3} = 333.33 \text{ K} = 60.33^\circ \text{C}$$

11.(C) For any solution, $\Delta S_{\text{mix}} = +\text{ve}$ and $\Delta G_{\text{mix}} = -\text{ve}$

12.(D) Colligative property. Adding a solute will decrease vapour pressure.

13.(C) $\pi = \frac{C}{M_w} \times 1000 \times R \times 300$

$$\text{Slope} = \frac{R \times 300}{M_w} \times 1000 = 5 \times 10^{-3}$$

14.(A) $2r = \frac{\sqrt{3}a}{4}$ or $r = \frac{\sqrt{3}a}{8} = \frac{1.732 \times 356}{8} = 77.07 \text{ pm}$

15.(D) The shortest distance between two Na^+ ion $= 1/2 \times \text{face-diagonal}$
 $= 1/2 \times \sqrt{2} \{2(r_{\text{Na}^+} + r_{\text{Cl}^-})\} = \sqrt{2}(95 + 181) \text{ pm} = 390.3 \text{ pm}$

16.(D) CsCl will show Schottky defect.

17.(B) Edge not covered by atom $= a - 2r$ & in FCC unit cell ; $r = \frac{a}{2\sqrt{2}}$

$$\text{Therefore, edge not covered} = a - 2 \times \frac{a}{2\sqrt{2}} \Rightarrow 0.293a$$

$$\text{Fraction of edge not covered} = 0.293$$

18.(C) Theory based

19.(A) Radius $= 1\text{\AA}$

$$\frac{\text{Packing Fraction}}{\text{Density}} = 0.1 \text{ cm}^3 / \text{gm}$$

$$\Rightarrow \frac{Z \times \frac{4}{3} \pi r^3 / a^3}{Z \times M / a^3 \times N_A} = 0.1 \Rightarrow \frac{4\pi r^3}{3 \times M} \times N_A = 0.1$$

$$M = \frac{4\pi r^3 \times N_A}{3 \times 0.1}$$

$$M = 8\pi$$

20.(C) For hexagonal unit cell $P.F. = \frac{3 \times \pi r^2}{6 \times \frac{\sqrt{3}}{4} (2r)^2} = \frac{\pi}{2\sqrt{3}}$

SECTION-2

21.(8) Fe_7S_8

Let No. of Fe^{+3} ions = x

$\therefore$ No. of Fe^{+2} ions = 7 - x

Total positive charge = Total Negative charge

$$3x + 2(7 - x) = 16$$

$$x = 2$$

So, No. of Fe^{+3} ions = 2

Three Fe^{+2} ions are replaced by two Fe^{+3} ions creating one cation vacancy.

$$\text{So percentage of cation vacancy} = \frac{1}{5+3} \times 100\% \quad \text{So Answer is 8.}$$

22.(500)

For simple cubic unit cell

$$2r = a \quad r = \text{radius of atom X}$$

If Radius of the largest atom that can be placed = R

$$\Rightarrow 2R = 366 \text{ pm}$$

$$2r + 2R = a\sqrt{3}$$

$$a + 366 = a\sqrt{3}$$

$$366 = a(\sqrt{3} - 1)$$

$$a = \frac{366}{0.732} = 500 \text{ pm}$$

$$= 5 \times 10^{-8} \text{ cm}$$

$$d = \frac{Zm}{a^3 N_A} \quad \text{Given } N_A = 6 \times 10^{23}$$

$$= \frac{1 \times 125}{125 \times 10^{-24} \times 6 \times 10^{23}} = \frac{10}{6}$$

$$d = \frac{5}{3} \text{ gm/cm}^3 \quad \text{So answer is } \frac{5 \times 300}{3} = 500$$

23.(177)

$$\pi_A = \pi_B$$

$$C_A RT = C_B RT \Rightarrow C_A = C_B$$

$$\Rightarrow \frac{n_A}{V_A} = \frac{n_B}{V_B} \Rightarrow \frac{0.73 / M_A}{250} = \frac{1.65 / M_B}{1000}$$

$$\Rightarrow \frac{M_A}{M_B} = 177 \times 10^{-2}$$

24.(777) $\Delta T_b = K_b \cdot m$

$$m = \left(\frac{\Delta T_b}{0.52} \right) = \frac{1.04}{0.52} = 2 \text{ mole/kg}$$

also $\frac{P^\circ - P_s}{P_s} = \frac{n_B}{n_A} = m \cdot M_A$

$$\frac{P^\circ}{P_s} - 1 = 2(18) \times 10^{-3}$$

$$\frac{P^\circ}{P_s} = 1 + 36 \times 10^{-3} = 1.036$$

$$P^\circ = 777 \text{ torr}$$

25.(90) $P = P_A^\circ x_A + P_B^\circ x_B$

$$600 = P_A^\circ \left(\frac{3}{3+2} \right) + P_B^\circ \left(\frac{2}{2+3} \right); 3P_A^\circ + 2P_B^\circ = 3000$$

$$630 = P_A^\circ \left(\frac{4.5}{4.5+2+0.5} \right) + P_B^\circ \left(\frac{2}{4.5+2+0.5} \right); 4.5P_A^\circ + 2P_B^\circ = 4410$$

$$1.5P_A^\circ = 1410; P_A^\circ = 940 \text{ \& } P_B^\circ = 90$$

Mathematics

SECTION-1

1.(D) For $x^2 - 4x + 5$ $D < 0$

$$\Rightarrow x^2 - 4x + 5 \in [1, \infty) \Rightarrow \sqrt{x^2 - 4x + 5} - 2 \in [-1, \infty) \Rightarrow \frac{1}{\sqrt{x^2 - 4x + 5} - 2} \in (-\infty, -1] \cup (0, \infty)$$

2.(C) $y = \frac{1}{x^2} = x^2 \Rightarrow x = 1, y = 1$

$$m_1 = \frac{dy}{dx} = -\frac{2}{x^3} = -2 ; \quad m_2 = \frac{dy}{dx} = 2x = 2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \frac{4}{3}$$

3.(D) $1 - \log_6(x^2 - 7x + 16) > 0$

$$\log_6(x^2 - 7x + 16) < 1$$

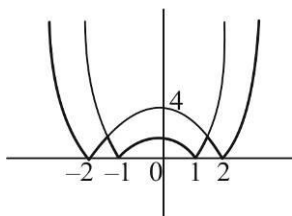
$$0 < x^2 - 7x + 16 < 6$$

$$x^2 - 7x + 10 < 0$$

$$(x - 2)(x - 5) < 0$$

$$x \in (2, 5)$$

4.(A) $\lim_{x \rightarrow -\infty} \frac{x^2 \tan\left(\frac{2}{x}\right)}{-x \sqrt{16 - \frac{1}{x} + \frac{1}{x^2}}} = -\frac{1}{2}$



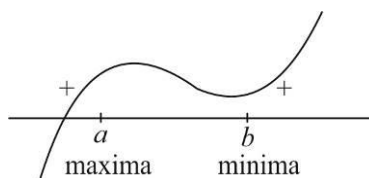
5.(B)

6.(B) $f'(x) = 6x^2 - 6(a+b)x + 6ab$

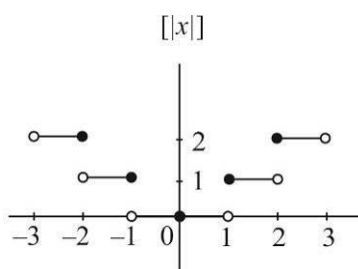
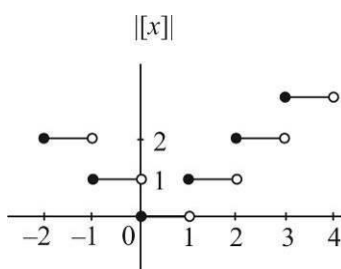
$$= 6(x - a)(x - b) = 0$$

$$x = a, b$$

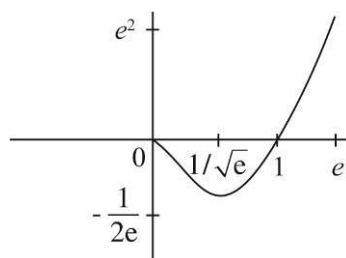
$$\text{If } a < b$$



7.(C)



8.(C) $f'(x) = 2x \ln x + x^2 \cdot \frac{1}{x}$
 $= 2x \ln x + x = x(2 \ln x + 1)$
 $f'(x) = 0$ for $x = e^{-1/2} = \frac{1}{\sqrt{e}}$
 Range $\left[-\frac{1}{2e}, e^2\right]$



9.(A) First make $f(x)$ continuous at $x = 2$
 $3a = 2b + 5$

For differentiability at 2, $a = 6b \Rightarrow b = \frac{5}{16}, a = \frac{15}{8}$

10.(C) $2^{1/4} = 16^{1/16}$

$g'(2^{1/4}) = \frac{1}{f'(16)}$

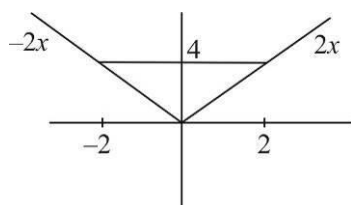
$f'(x) = x^{1/x} \left(\frac{1 - \ln x}{x^2} \right); f'(16) = 16^{1/16} \left(\frac{1 - 4 \ln 2}{2^8} \right); \frac{1}{f'(16)} = \frac{2^{31/4}}{1 - 4 \ln 2}$

11.(A) $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$

$\lim_{x \rightarrow 0} \frac{\left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right)^3 - \left(x^3 + \frac{x^9}{3} + \frac{2}{15}x^{15} + \dots \right)}{x^5}; \lim_{x \rightarrow 0} \frac{x^3 + 3x^2 \left(\frac{x^3}{3} + \frac{2}{15}x^5 + \dots \right) - x^3}{x^5} = 1$

12.(A) $2f(x-1) = f(x-3) + f(x+1)$

$2|x| = |x-2| + |x+2|$



13.(C) $f(0) = 0$ (by taking $x = y = 0$)

$y = -x$

$f(0) = f(x) + f(-x) = 0$

14.(A) By LMVT $\exists c \in (0, x) \forall x \in (0, 4]$ such that

$f'(c) = \frac{f(x) - f(0)}{x}$

$-2 \leq \frac{f(x) - f(0)}{x} \leq 2$

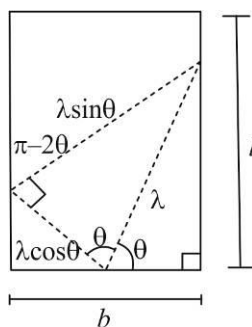
$-2x \leq f(x) \leq 2x \forall x \in (0, 4]$

15.(B) $\lambda \sin \theta \sin 2\theta = b$

$$A = \frac{1}{2} \lambda^2 \cos \theta \sin \theta$$

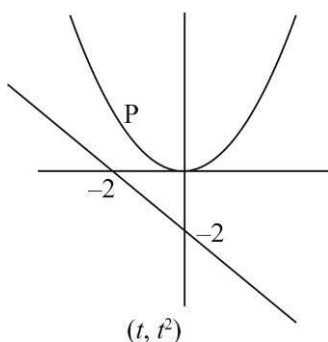
$$= \frac{b^2}{8 \sin^3 \theta \cos \theta}$$

Minimum for $\theta = 60^\circ$



16.(C) $P(t, t^2)$

$$\frac{dy}{dx} = 2t = -1 \Rightarrow -\frac{1}{2} \quad ; \quad P\left(-\frac{1}{2}, \frac{1}{4}\right) \text{ its distance from } x + y + 2 = 0 \text{ is } \left| \frac{-\frac{1}{2} + \frac{1}{4} + 2}{\sqrt{2}} \right| = \frac{7}{4\sqrt{2}}$$



17.(A) $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) \cdot f(h) - f(x)}{h}$

$$= f(x) \lim_{h \rightarrow 0} \frac{hg(h) \ln 3}{h} = f(x) \ln 3 \Rightarrow f(x) = 3^x$$

18.(B) $f'(x) = 24x^2 + 8x + 2k = 2(12x^2 + 4x + k)$

$$D \leq 0$$

$$16 - 48k \leq 0$$

$$\frac{1}{3} \leq k$$

19.(D) $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$

$$= (2at + 2b)(2ax + 2b) = 4(at + b)(ax + b)$$

$$\frac{d^2y}{dx^2} = 4((at + b)a + (ax + b)a \cdot (2ax + 2b))$$

$$= 4a((at + b) + 2(ax + b)^2)$$

$$\frac{d^3y}{dx^3} = 4a(a(2ax + 2b) + 4(ax + b) \cdot a)$$

$$= 24a^2(ax + b)$$

20.(A) $\frac{\sin x}{x}$ & $\frac{x}{\tan x}$ are even functions

For $x > 0$ $\sin x < x < \tan x$
 $\frac{\sin x}{x} < 1$ & $\frac{x}{\tan x} < 1$

$$\Rightarrow \lim_{x \rightarrow 0} \left[\frac{3 \sin x}{x} \right] = 2 \text{ \& } \lim_{x \rightarrow 0} \left[\frac{4x}{\tan x} \right] = 3$$

SECTION-2

21.(7) $f(x) = (x+2)^2 + \frac{1}{x^2} - 4x, x > 0, x \neq 1$

$$f(x) = x^2 + 4 + 4x + \frac{1}{x^2} - 4x, x > 0, x \neq 1$$

$$= x^2 + \frac{1}{x^2} + 4 > 6$$

22.(1) $\lim_{x \rightarrow 0} \frac{(9^x - 1)}{x} \times \frac{(3^x - 1)}{x} \times \frac{\cos x - e^x}{x^{2n-1}}$

$$= \ln 9 \times \ln 3 \times \lim_{x \rightarrow 0} \frac{\left(1 - \frac{x^2}{2!} + \dots\right) - \left(1 + x + \frac{x^2}{2!}\right)}{x^{2n-1}}$$

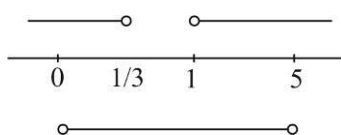
exist if $n = 1$

23.(6) $\text{LHD} = \lim_{h \rightarrow 0} \frac{[-5-h]\sin(\pi(-5-h)) - [-5]\sin(-5\pi)}{-h} = \lim_{h \rightarrow 0} \frac{6\sin(5\pi + \pi h)}{-h} = \lim_{h \rightarrow 0} \frac{6\sin \pi h}{h} = 6\pi$

24.(3) $f(2a^2 + a + 1) > f(3a^2 - 4a + 1)$

$$\Rightarrow 2a^2 + a + 1 > 3a^2 - 4a + 1 > 0 \Rightarrow a^2 - 5a < 0 \text{ \& } (3a-1)(a-1) > 0$$

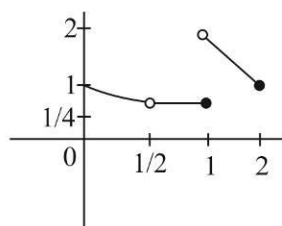
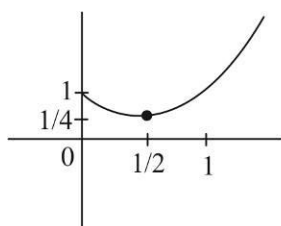
$$\Rightarrow a(a-5) < 0 \text{ \& } (3a-1)(a-1) > 0$$



$$a \in \left(0, \frac{1}{3}\right) \cup (1, 5)$$

25.(5) $f'(x) = 12x^2 - 2x - 2$

$$= 2(6x^2 - x - 1) = 2(3x+1)(2x-1)$$



$$2 \left(g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right) \right) = 2 \left(f\left(\frac{1}{4}\right) + \frac{1}{4} + 3 - \frac{5}{4} \right) = 5$$